Multi-Objective Evolutionary Algorithm for PET Image Reconstruction

DA244 Course Project

MOEAP & R-MOEAP Framework

Multi-Objective Optimisation of Positron Emission Tomography Reconstruction

April 15, 2026

1 Introduction

Positron Emission Tomography (PET) image reconstruction is an ill-posed inverse problem: from a measured sinogram $\mathbf{y} \in \mathbb{R}^M$ one must recover an activity map $\mathbf{x} \in \mathbb{R}_+^N$. Classical single-objective approaches (MLEM, MAP) optimise one scalar criterion, forcing the practitioner to make prior trade-off decisions that profoundly affect diagnostic utility.

We present **MOEAP** (Multi-Objective Evolutionary Algorithm for PET) and its reference-point extension **R-MOEAP**, which jointly optimise three competing image-quality surrogates:

1. Poisson log-likelihood (ℓ_{PET}) — data fidelity.
2. CNN-predicted noise-signal ratio (NSNR) — perceptual noise quality.
3. CNN-predicted inverse RMSE (1/RMSE) — accuracy vs. a reference.

The result is a *Pareto front* of reconstructions, giving the clinician explicit control over the bias–variance trade-off.

2 PET Forward Model

2.1 System Matrix

The discretised 2-D parallel-beam forward model is

$$\bar{\mathbf{y}} = \mathbf{A}\mathbf{x} + \mathbf{s}, \tag{1}$$

where $\mathbf{A} \in \mathbb{R}_+^{M \times N}$ is the system matrix (one row per detector–angle pair), $\mathbf{x} \in \mathbb{R}_+^N$ is the unknown tracer activity image ($N = H \times W$, here 64^2), and $\mathbf{s} \in \mathbb{R}_+^M$ models scatter and random coincidences. $\mathbf{A}$ is constructed by ray-driven line integration using a Bresenham-like approach and row-normalised so that $\sum_j A_{ij} = 1$.

2.2 Poisson Statistics

Each sinogram bin i follows an independent Poisson distribution:

$$y_i \sim \mathrm{Poisson}(\bar{y}_i)\,, \quad \bar{y}_i = [\mathbf{A}\mathbf{x}]_i + s_i. \tag{2}$$

The Poisson log-likelihood (data-fidelity objective) is

$$\ell_{\mathrm{PET}}(\mathbf{x}) = \sum_{i=1}^{M} \Big[-\bar{y}_i + y_i \ln \bar{y}_i \Big]. \tag{3}$$

3 Multi-Objective Problem Formulation

3.1 Objective Space

Define the objective vector

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ f_3(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \ell_{\mathrm{PET}}(\mathbf{x}) \\ \widehat{\mathrm{NSNR}}(\mathbf{x}) \\ (\widehat{1/\mathrm{RMSE}})(\mathbf{x}) \end{bmatrix}, \tag{4}$$

where $\hat{\cdot}$ denotes CNN-predicted surrogates (Section 4). All objectives are to be **maximised**.

3.2 Pareto Dominance

Solution $\mathbf{x}^{(a)}$ *dominates* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \succ \mathbf{x}^{(b)}$, if and only if

$$f_m(\mathbf{x}^{(a)}) \geq f_m(\mathbf{x}^{(b)}) \quad \forall m, \tag{5}$$

$$f_m(\mathbf{x}^{(a)}) > f_m(\mathbf{x}^{(b)}) \quad \text{for some } m. \tag{6}$$

The *Pareto-optimal set* $\mathcal{P}^*$ consists of all solutions not dominated by any other feasible $\mathbf{x}$.

4 CNN Objective Surrogates

Computing ℓ_{PET} analytically is cheap, but evaluating RMSE and NSNR requires a reference scan unavailable at inference time. We train three separate CNN regressors $\hat{f}_m : \mathbb{R}^{H\times W} \to \mathbb{R}$ on a synthetic dataset of 10 000 PET reconstructions with known ground-truth labels.

4.1 Architecture

Each model follows a four-block encoder:

$$\text{Block}_k : \ \text{Conv}(3{\times}3) \to \text{AvgPool}(2) \to \text{BN} \to \text{ReLU}, \tag{7}$$

with channel depths $[1, 8, 16, 32, 32]$. After four pooling stages a 64×64 input is reduced to a 4×4 feature map (1024 values at depth 32), which is flattened and passed through Dropout(0.2) and a linear head to scalar output.

4.2 Training Protocol

Models are optimised with

$$\min_{\boldsymbol{\theta}_m} \ \mathcal{L}_m = \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{x}, y_m)\in\mathcal{D}} \big(\hat{f}_m(\mathbf{x}; \boldsymbol{\theta}_m) - y_m\big)^2, \tag{8}$$

using SGD (momentum 0.9, weight decay 10^{-4}), cosine annealing LR over 60 epochs, and early stopping with patience 15.

5 MOEAP: Algorithm

MOEAP extends NSGA-II with a PET-specific directed mutation operator. The main loop is given in Algorithm 1.

5.1 Initialisation

The initial population $P_0 = \{\mathbf{x}_0^{(i)}\}_{i=1}^{N_p}$ is generated by perturbing a filtered back-projection (FBP) estimate:

$$\mathbf{x}_0^{(i)} = \text{clip}\Big(\mathbf{x}_{\text{FBP}} + \boldsymbol{\epsilon}^{(i)},\ 0, \infty\Big), \quad \boldsymbol{\epsilon}^{(i)} \sim \mathcal{U}(-0.1, 0.1), \tag{9}$$

where $\mathbf{x}_{\text{FBP}} = \mathbf{A}^\top \mathbf{y} / \max(\mathbf{A}^\top \mathbf{y})$.

5.2 Non-Dominated Sorting (Fast Sort)

Given objective matrix $\mathbf{F} \in \mathbb{R}^{N_p \times M}$, the population is partitioned into fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$ by the classical NSGA-II fast non-dominated sort, running in $O(N_p^2 M)$. Individuals in $\mathcal{F}_1$ are non-dominated; $\mathcal{F}_k$ consists of solutions dominated only by members of $\mathcal{F}_1, \ldots, \mathcal{F}_{k-1}$.

5.3 Crowding Distance

Within front $\mathcal{F}_k$ the crowding distance of individual i is

$$d^{(i)} = \sum_{m=1}^{M} \frac{f_m\big(\mathbf{x}^{(i+1)}\big) - f_m\big(\mathbf{x}^{(i-1)}\big)}{f_m^{\max} - f_m^{\min}}, \tag{10}$$

where neighbours are sorted along objective m. Boundary individuals receive $d = \infty$.

5.4 Tournament Selection

Binary tournament selects the parent with the lower front rank; ties are broken in favour of the larger crowding distance (preserving diversity).

5.5 Simulated Binary Crossover (SBX)

Operating on flattened image vectors $\mathbf{p}_1, \mathbf{p}_2 \in \mathbb{R}^N$, SBX with distribution index $\eta_c = 20$ and crossover probability $p_c = 0.95$ produces offspring:

$$\mathbf{c}_1 = \tfrac{1}{2}[(1 + \boldsymbol{\beta})\mathbf{p}_1 + (1 - \boldsymbol{\beta})\mathbf{p}_2]\,, \tag{11}$$

$$\mathbf{c}_2 = \tfrac{1}{2}[(1 - \boldsymbol{\beta})\mathbf{p}_1 + (1 + \boldsymbol{\beta})\mathbf{p}_2]\,, \tag{12}$$

where elementwise β_j is drawn as

$$\beta_j = \begin{cases} (2u_j)^{1/(\eta_c+1)} & u_j \le 0.5, \\ \left(\frac{1}{2(1-u_j)}\right)^{1/(\eta_c+1)} & u_j > 0.5, \end{cases} \quad u_j \sim \mathcal{U}(0, 1). \tag{13}$$

5.6 Directed Mutation

Standard Gaussian mutation is replaced by a PET-aware gradient step:

$$\mathbf{x}_{\text{mut}} = \text{clip}\big(\mathbf{x} + \sigma\, \nabla_{\mathbf{x}} \ell_{\text{PET}}(\mathbf{x}) + \sigma \cdot 0.1\ \boldsymbol{\eta},\ 0, \infty\big), \tag{14}$$

where $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\sigma = 0.01$. The gradient of the Poisson log-likelihood is

$$\nabla_{\mathbf{x}} \ell_{\text{PET}} = \mathbf{A}^\top \left(\frac{\mathbf{y}}{\bar{\mathbf{y}}} - \mathbf{1} \right), \tag{15}$$

with division applied elementwise. This biases mutations toward sinogram-consistent solutions while preserving non-negativity.

5.7 Environmental Selection

Offspring Q_t ($|Q_t| = N_p$) are combined with parents P_t to form $R_t = P_t \cup Q_t$ ($|R_t| = 2N_p$). The next generation P_{t+1} is assembled greedily: fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$ are admitted in order until the population cap N_p is reached; the last admitted front is truncated by decreasing crowding distance.

Algorithm 1 MOEAP Main Loop

Require: $\mathbf{y}$, $\mathbf{A}$, CNN models, N_p, T

$P_0 \leftarrow$ FBP-seeded population (N_p images)
Evaluate $\mathbf{F}_0 \leftarrow \mathbf{f}(P_0)$
$\{\mathcal{F}_k\} \leftarrow \textsc{FastSort}(\mathbf{F}_0)$
for $t = 1, \ldots, T$ **do**
  parents $\leftarrow \textsc{BinaryTournament}(P_{t-1}, \mathbf{F}_{t-1})$
  $Q_t \leftarrow$ SBX(parents) $\rightarrow$ $\textsc{DirectedMutate}$ (Eq. 14)
  Evaluate $\mathbf{F}_Q \leftarrow \mathbf{f}(Q_t)$
  $R_t \leftarrow P_{t-1} \cup Q_t$; $\mathbf{F}_R \leftarrow [\mathbf{F}_{t-1}; \mathbf{F}_Q]$
  $\{\mathcal{F}_k\} \leftarrow \textsc{FastSort}(\mathbf{F}_R)$
  $P_t \leftarrow \textsc{FillByRank}(\{\mathcal{F}_k\}, N_p)$
  $\mathbf{F}_t \leftarrow \mathbf{f}(P_t)$
end for
return P_T, $\mathbf{F}_T$, $\{\mathcal{F}_k\}$

6 R-MOEAP: Reference-Point Extension

R-MOEAP biases the search towards a user-supplied reference point $\mathbf{r} \in \mathbb{R}^M$ in objective space, analogous to the aspiration vector in PBEA and MOEA/D.

6.1 Proximity Score

For individual i on the critical front, define

$$\rho^{(i)} = \min_k \left\| \mathbf{f}(\mathbf{x}^{(i)}) - \mathbf{r}^{(k)} \right\|_2, \tag{16}$$

where $\{\mathbf{r}^{(k)}\}$ is the set of reference points. Individuals are ranked by ascending ρ (smaller = closer to desired region).

6.2 ε-Clearance Diversity

To prevent the selected subset from collapsing to the same point, a clearance radius ε enforces:

$$\left\| \mathbf{f}(\mathbf{x}^{(i)}) - \mathbf{f}(\mathbf{x}^{(j)}) \right\|_2 \geq \varepsilon \quad \forall\, i \neq j \text{ selected.} \tag{17}$$

The greedy selection picks candidates in order of $\rho^{(i)}$ and rejects any that violate Eq. (17). If the quota n_{needed} is not filled, the constraint is relaxed and candidates are added greedily.

7 KKTPM Convergence Measure

7.1 Motivation

Hypervolume and generational distance measure quality of the approximate front but do not directly indicate proximity to Pareto optimality. The *Karush–Kuhn–Tucker Proximity Measure* (KKTPM) fills this role by quantifying how well the KKT necessary conditions for multi-objective optimality are satisfied.

7.2 KKT Conditions for the MOP

At a Pareto-stationary point $\mathbf{x}^*$, the KKT conditions require that there exist Lagrange multipliers $\boldsymbol{\lambda} \in \mathbb{R}^M$ such that

$$\sum_{m=1}^{M} \lambda_m \, \nabla_{\mathbf{x}} f_m(\mathbf{x}^*) = \mathbf{0}, \tag{18}$$

$$\lambda_m \geq 0 \quad \forall m, \qquad \sum_{m=1}^{M} \lambda_m = 1. \tag{19}$$

7.3 KKTPM Definition

For a candidate $\mathbf{x}$, KKTPM is defined as

$$\text{KKTPM}(\mathbf{x}) = \min_{\boldsymbol{\lambda} \in \Delta^{M-1}} \left\| \sum_{m=1}^{M} \lambda_m \, \nabla_{\mathbf{x}} f_m(\mathbf{x}) \right\|_2, \tag{20}$$

where $\Delta^{M-1} = \{\boldsymbol{\lambda} : \lambda_m \geq 0,\ \sum_m \lambda_m = 1\}$ is the probability simplex. $\text{KKTPM}(\mathbf{x}) = 0$ iff $\mathbf{x}$ is Pareto-stationary.

7.4 Numerical Approximation

Gradients are approximated by central finite differences. For computational tractability, a random subsample of $K_s = 500$ pixel indices $\mathcal{S} \subset \{1, \ldots, N\}$ is used:

$$[\nabla_{\mathbf{x}} f_m(\mathbf{x})]_j \approx \frac{f_m(\mathbf{x} + h\mathbf{e}_j) - f_m(\mathbf{x} - h\mathbf{e}_j)}{2h}, \quad j \in \mathcal{S}, \tag{21}$$

with $h = 10^{-4}$. Let $\mathbf{G} \in \mathbb{R}^{M \times K_s}$ collect these columns.

The inner minimisation in Eq. (20) over the simplex becomes

$$\min_{\boldsymbol{\lambda} \in \Delta^{M-1}} \boldsymbol{\lambda}^\top \mathbf{G}\mathbf{G}^\top \boldsymbol{\lambda}. \tag{22}$$

This is solved by projected gradient descent:

$$\boldsymbol{\lambda}^{(\tau+1)} = \Pi_\Delta \left(\boldsymbol{\lambda}^{(\tau)} - \alpha \, \mathbf{G}\mathbf{G}^\top \boldsymbol{\lambda}^{(\tau)} \right), \tag{23}$$

with step size $\alpha = 0.01$ and 200 iterations. The projection Π_Δ onto the probability simplex uses the

$O(M \log M)$ closed-form sort algorithm [2]:

$$u_k = \mathrm{sort}(v)_k \text{ (descending)}, \tag{24}$$

$$\rho^* = \max\left\{k : u_k - \tfrac{1}{k}\left(\textstyle\sum_{j \le k} u_j - 1\right) > 0\right\}, \tag{25}$$

$$\theta^* = \tfrac{1}{\rho^*}\left(\textstyle\sum_{j \le \rho^*} u_j - 1\right), \tag{26}$$

$$[\Pi_\Delta(\mathbf{v})]_k = \max(v_k - \theta^*, 0). \tag{27}$$

7.5 Normalised KKTPM

The final scalar is normalised to $[0, 1]$:

$$\widetilde{\mathrm{KKTPM}}(\mathbf{x}) = \frac{\left\|\mathbf{G}^\top \boldsymbol{\lambda}^*\right\|_2}{\|\mathbf{G}\|_F + \epsilon}, \tag{28}$$

where $\|\cdot\|_F$ is the Frobenius norm. Median KKTPM over the first Pareto front is tracked per generation as the primary convergence criterion; the algorithm is deemed converged when this value stagnates.

8 Baseline Comparators

8.1 MLEM

The standard Expectation-Maximisation (ML-EM) update is

$$\mathbf{x}^{(k+1)} = \frac{\mathbf{x}^{(k)}}{\mathbf{A}^\top \mathbf{1}} \odot \mathbf{A}^\top \left(\frac{\mathbf{y}}{\mathbf{A}\mathbf{x}^{(k)} + \mathbf{s}}\right), \tag{29}$$

run for 200 iterations from $\mathbf{x}^{(0)} = \mathbf{1}$. Post-smoothing sweeps Gaussian FWHM $\in \{2, 2.5, \ldots, 7.5\}$ px, yielding a 1-D parameter family for comparison.

8.2 MAP with Quadratic Penalty

The MAP objective is

$$\max_{\mathbf{x} \ge \mathbf{0}} \; \ell_{\mathrm{PET}}(\mathbf{x}) - \beta \|\mathbf{x}\|_2^2, \tag{30}$$

optimised by gradient ascent with step $\alpha_j = ([\mathbf{A}^\top \mathbf{1}]_j + 2\beta)^{-1}$ and projected onto the non-negative orthant. Six regularisation strengths $\beta \in \{0.1, 0.5, 1, 5, 10, 50\}$ yield another 1-D comparison curve.

9 Experimental Setup

Dataset generation produces 10 000 synthetic 2-D phantoms with random organ backgrounds and inserted spherical lesions (contrast 3×, radius 5 px). Sinograms are simulated via Eq. (1) with Poisson noise. Labels (RMSE, NSNR, FWHM) are computed against the ground-truth activity map.

Table 1: Hyperparameters used across all experiments.

Component	Parameter	Value
PET model	Image size $H \times W$	64×64
	Projection angles	60
	Total counts n_e	8×10^4
	Scatter fraction	10%
MOEAP	Population N_p	50
	Generations T	100
	SBX η_c	20
	SBX p_c	0.95
	Mutation strength σ	0.01
R-MOEAP	ε-clearance	0.05
	Reference point	Mean of MOEAP front
CNN	Epochs	60
	Batch size	64
	Optimizer	SGD + cosine LR
	Training set size	8 000
KKTPM	FD step h	10^{-4}
	Sub-sample K_s	500
	PGD iterations	200
	PGD step α	0.01

10 Results Overview

The Pareto front obtained by MOEAP is expected to strictly dominate the 1-D curves traced by MLEM and MAP baselines in the (ℓ_{PET}, NSNR)-plane, demonstrating that no single-parameter family can explore the full trade-off surface.

R-MOEAP concentrates solutions near the mean MOEAP front point, reducing front spread but increasing density around the clinically preferred region.

KKTPM median on the first front should decrease monotonically during evolution; stagnation below 10^{-2} serves as the convergence certificate.

11 Discussion

Directed mutation vs. random mutation. Replacing isotropic Gaussian perturbations with the Poisson gradient (Eq. 14) exploits domain structure: offspring are always pushed toward sinogram-consistent images, dramatically reducing the fraction of infeasible (physically inconsistent) individuals per generation.

CNN surrogates as objectives. Training three independent regressors allows unconstrained exploration of the objective space. The alternative of computing RMSE against a fixed reference scan would couple

the optimisation to a specific patient, whereas CNN predictions generalise across phantoms.

KKTPM vs. hypervolume. Hypervolume measures the volume of dominated space and is sensitive to the choice of reference point. KKTPM directly interrogates the stationarity conditions and is reference-point free, making it a complementary and more theoretically grounded indicator for multi-objective convergence.

12 Conclusion

We described MOEAP and R-MOEAP, evolutionary algorithms that reformulate PET reconstruction as a multi-objective problem. Key contributions are: (i) a CNN-based surrogate objective system trained once and reused; (ii) PET-aware directed mutation that enforces Poisson consistency; (iii) reference-point guidance in R-MOEAP for clinical specificity; (iv) KKTPM as a principled, gradient-based stopping criterion.

Together these provide clinicians with an explicit Pareto front of reconstructions rather than a single opaque hyper-parameter choice.

References

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